COMPLEX COMPUTING PROBLEM

<u>COURSE:</u> **CLOUD COMPUTING & VIRTULIZATION**

<u>FACULTY NAME:</u> **SIR ASFAND YAR**

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<u>CLASS TIMING</u>: SUNDAY(2:30 – 05:30 Pm)

Introduction:

Scholarship management is an important process in educational institutions. Traditional methods often involve paperwork, manual verification, and inefficient communication between students and administrators.

To solve these issues, a web-based Scholarship Management Portal was developed and deployed on AWS cloud infrastructure. The portal enables online scholarship applications while utilizing cloud services for scalability, availability, security, monitoring, and storage.

Objectives

- Develop an online scholarship management system.
- Deploy the application on AWS cloud infrastructure.
- Implement secure database connectivity.
- Provide centralized application management.
- Monitor cloud resources using AWS CloudWatch.
- Store documents using Amazon S3.
- Demonstrate Infrastructure as Code using Terraform.

Literature Review / Related Work:

Traditional Scholarship Systems

Conventional scholarship systems rely heavily on manual documentation, resulting in processing delays and increased administrative workload.

Cloud-Based Education Systems

Modern educational platforms leverage cloud computing to provide:

- High availability
- Scalability
- Reduced operational costs
- Automated backups
- Secure data storage

AWS in Educational Applications

AWS provides various services suitable for educational systems:

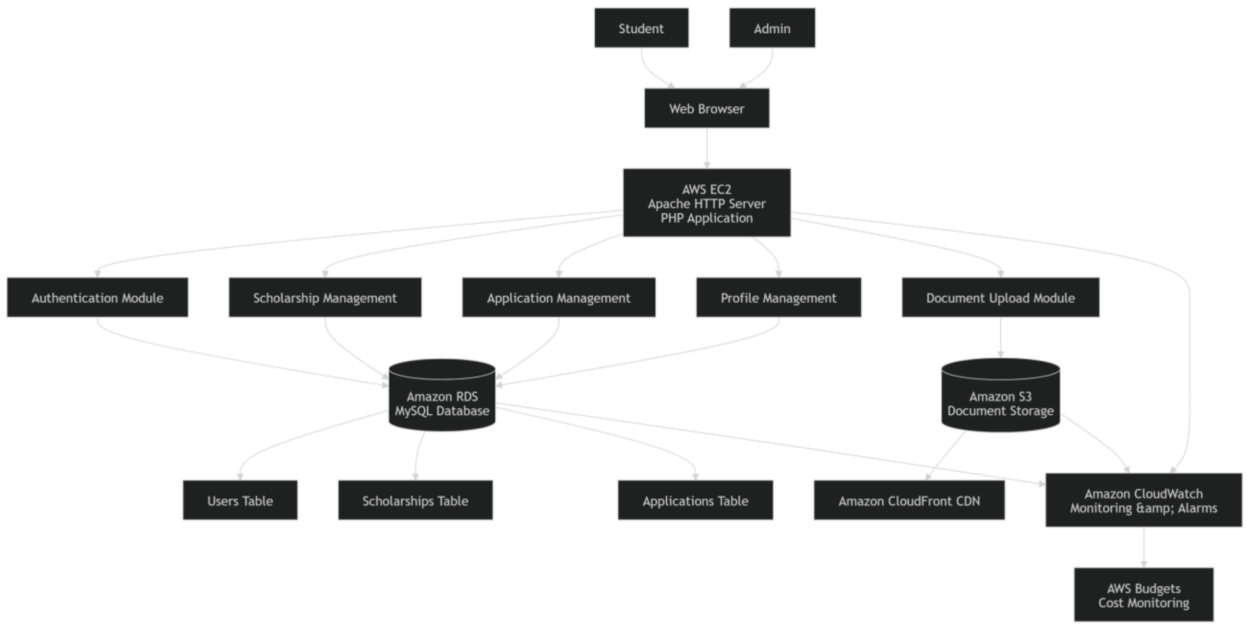
- EC2 for hosting applications
- RDS for managed databases
- S3 for document storage

- CloudFront for content delivery
- CloudWatch for monitoring
- Terraform for infrastructure automation

These services collectively enable the development of reliable cloud-native applications.

System Design and Architecture

Overall Architecture



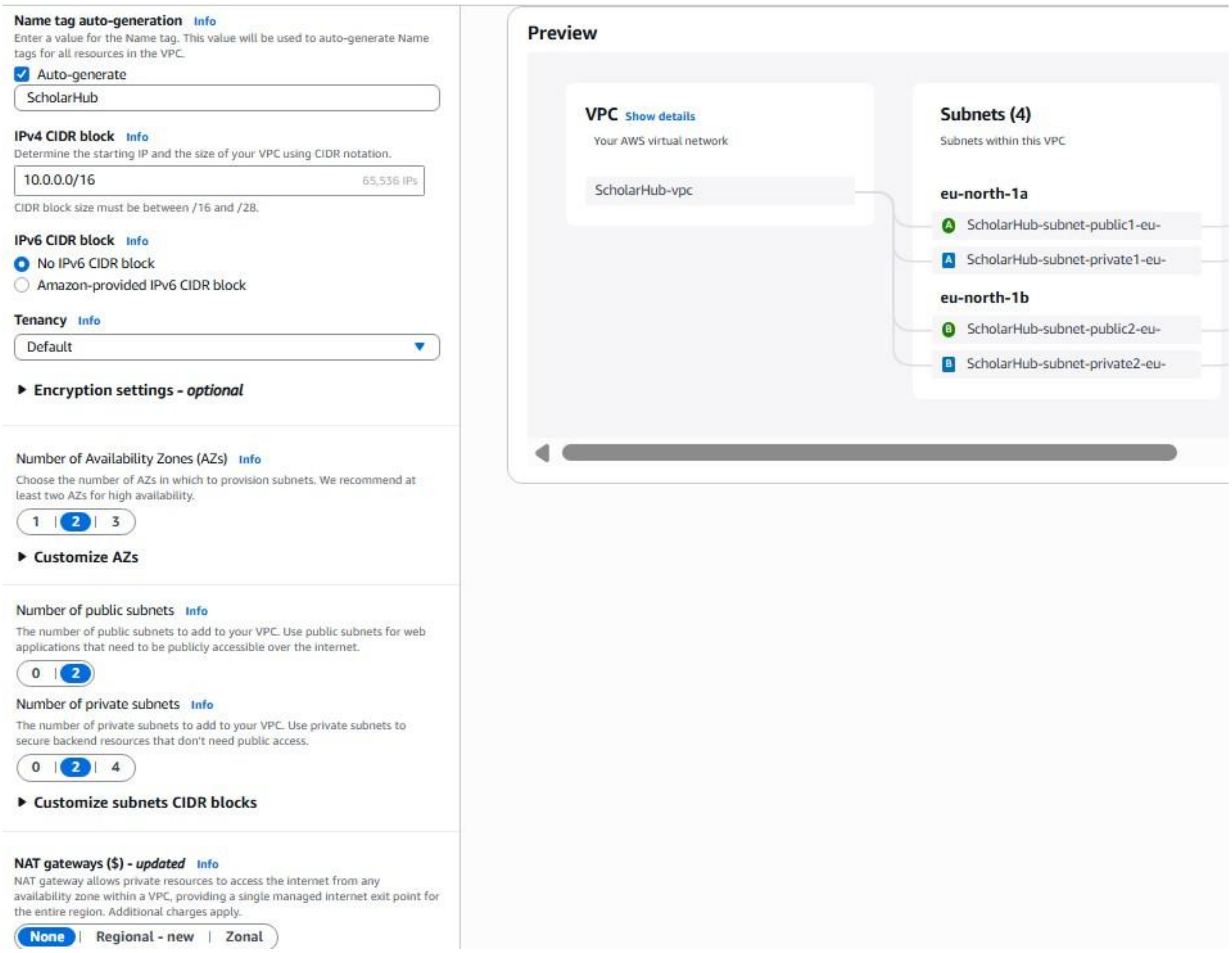
Architecture Components

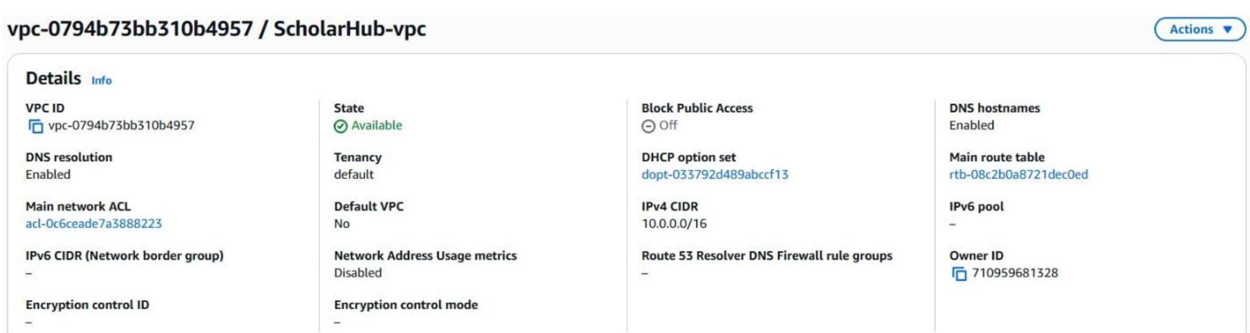
1. Student
2. Admin
3. Web Browser
4. ScholarHub Portal (PHP Application)
5. Authentication Module
6. Scholarship Management Module
7. Application Management Module
8. Profile Management Module
9. Document Upload Module
10. MySQL Database
11. Users Table
12. Scholarships Table
13. Applications Table
14. Uploads Folder

AWS Infrastructure:

Virtual Private Cloud (VPC)

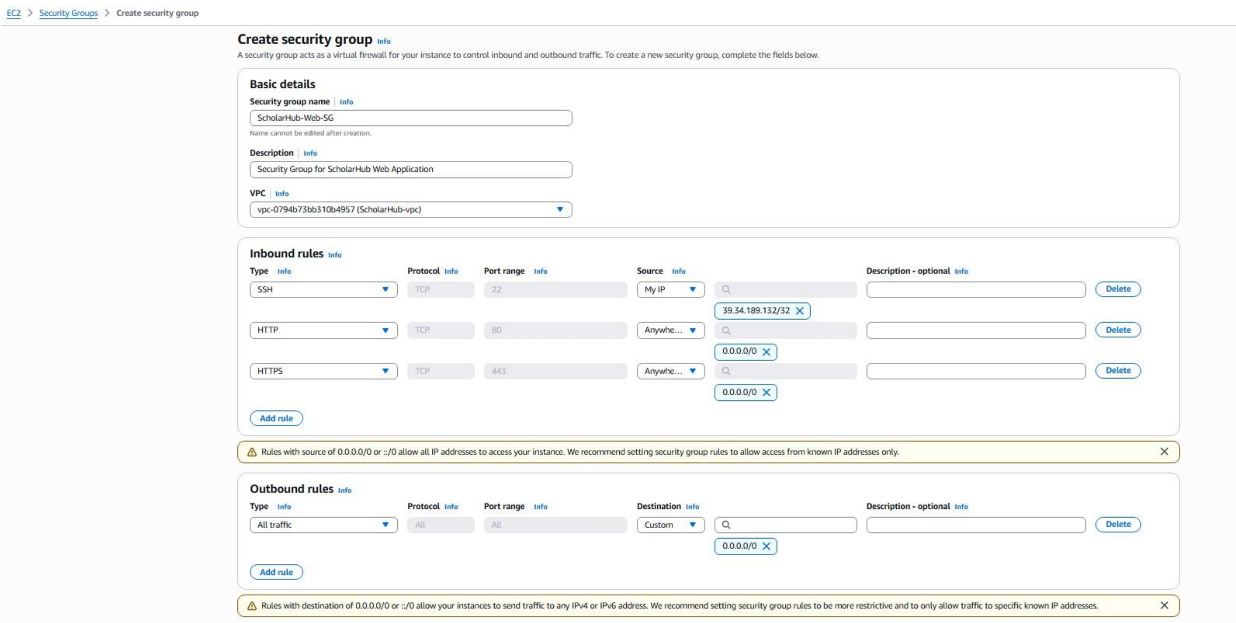
A dedicated VPC was configured to isolate project resources and provide secure communication.





Security Groups

Security Groups were configured to control inbound and outbound traffic.



Ports:

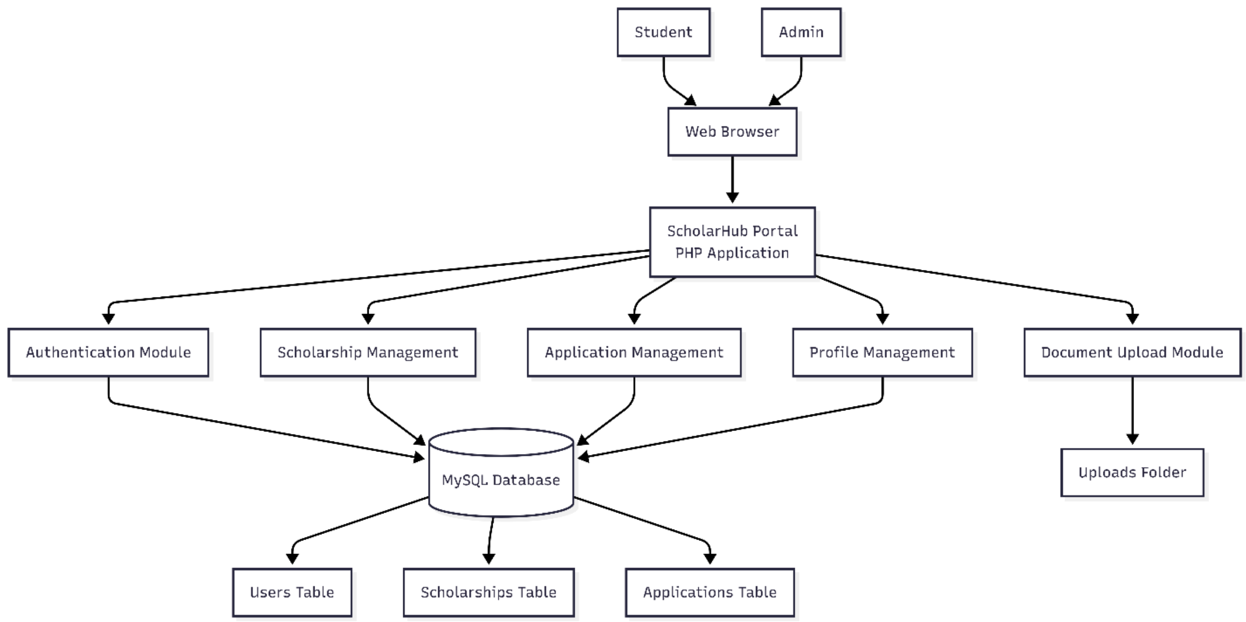
- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)
- 3306 (MySQL)

Methodology:

Phase 1: Local Development

The application was initially developed in XAMPP using:

- PHP
- HTML
- CSS
- JavaScript
- MySQL



Database Design:

Database tables:

Users

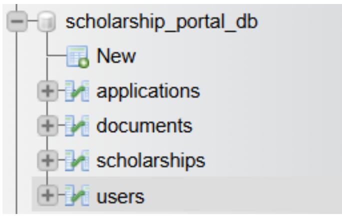
- id
- name
- email
- password
- role

Scholarships

- scholarship_id
- title
- description
- amount

Applications

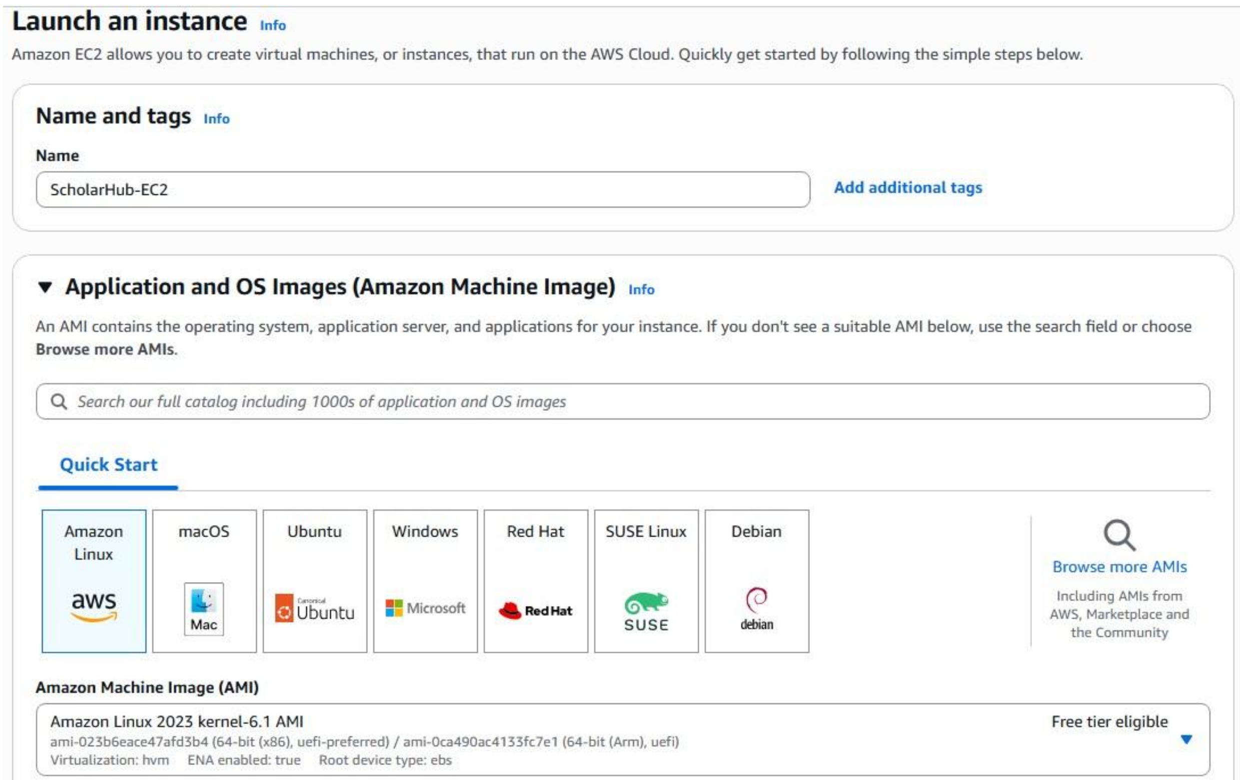
- application_id
- scholarship_id
- user_id
- document

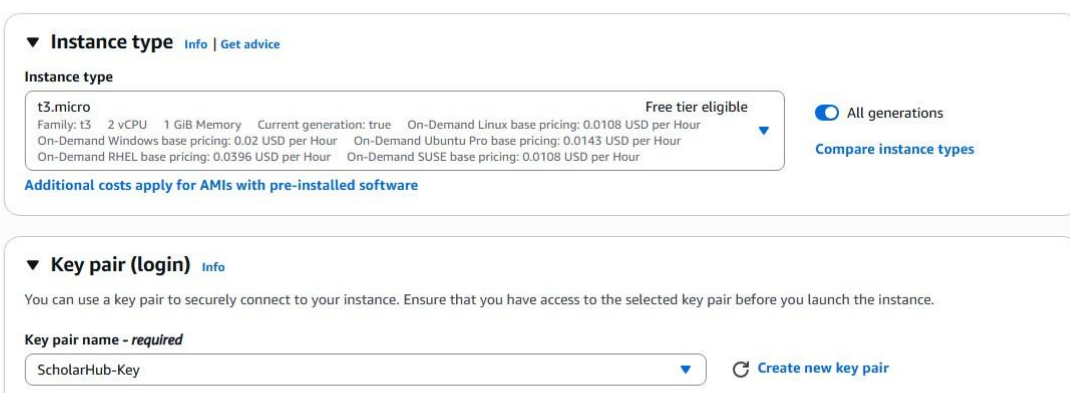


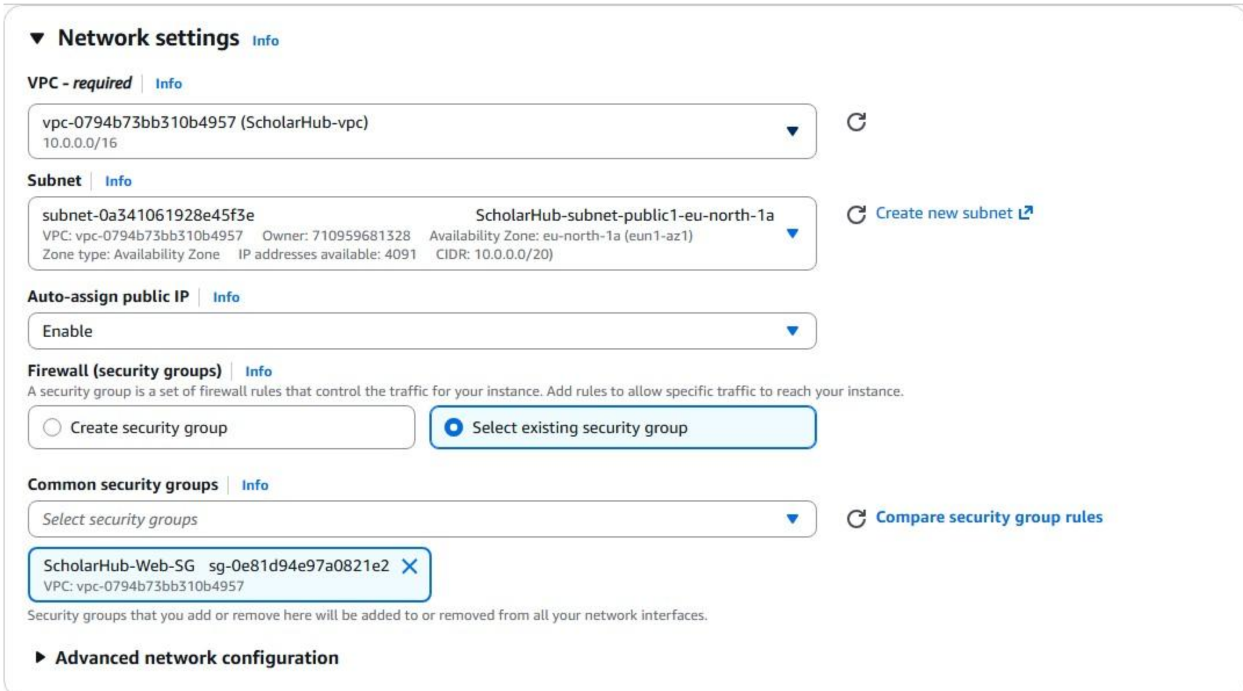
AWS Deployment:

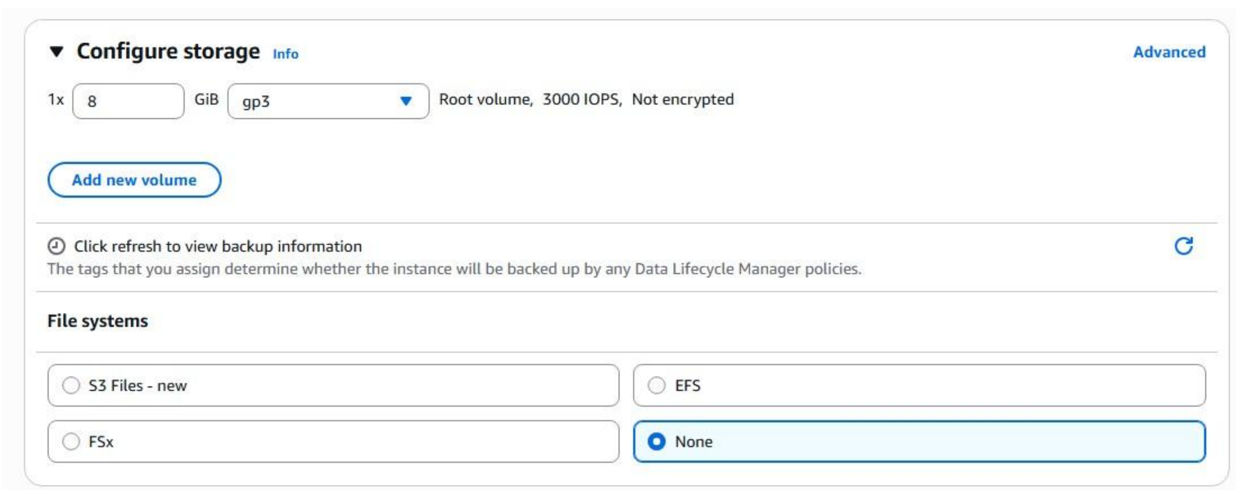
EC2 Configuration

Amazon EC2 instance was launched and configured as the application server.



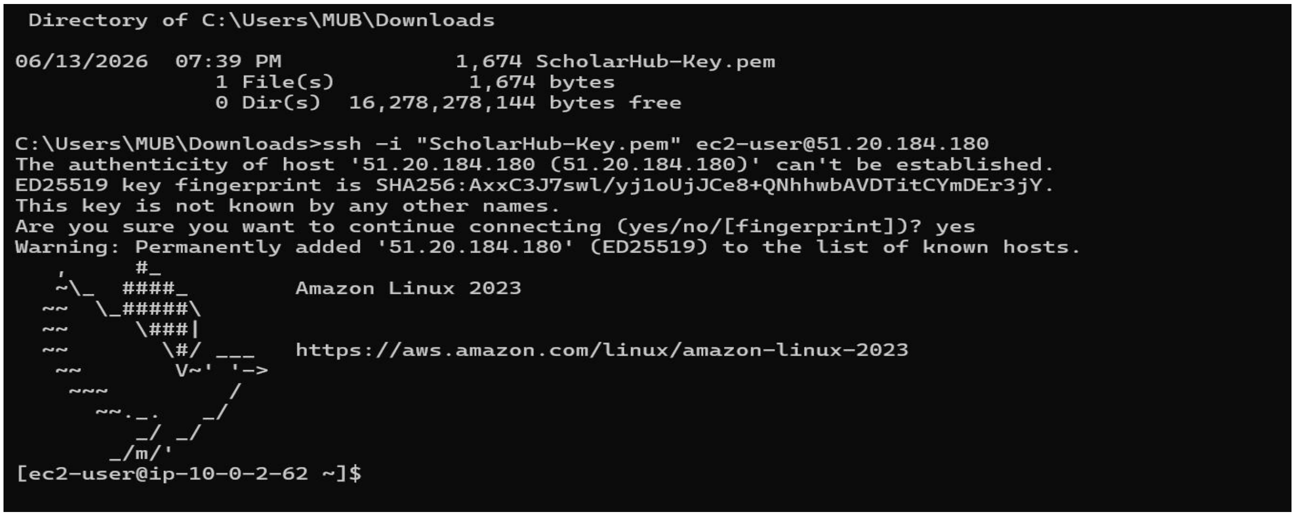






SSH Connectivity:

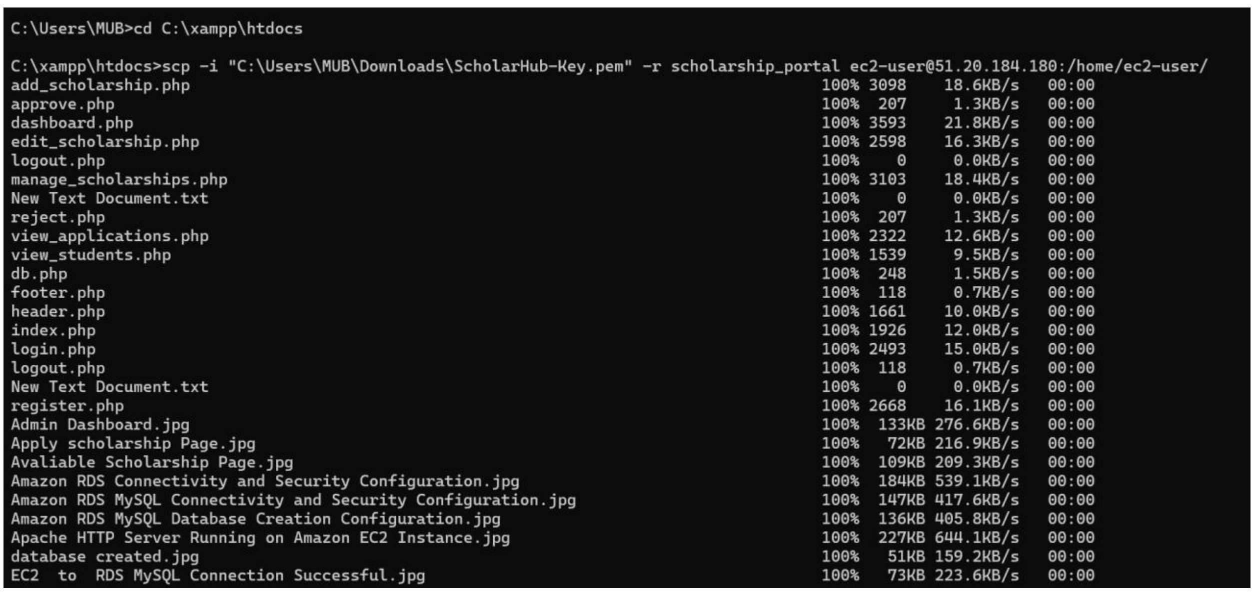
Remote administration was performed through SSH.



```
 Directory of C:\Users\MUB\Downloads

06/13/2026  07:39 PM             1,674 ScholarHub-Key.pem
               1 File(s)          1,674 bytes
               0 Dir(s)  16,278,278,144 bytes free

C:\Users\MUB\Downloads>ssh -i "ScholarHub-Key.pem" ec2-user@51.20.184.180
The authenticity of host '51.20.184.180 (51.20.184.180)' can't be established.
ED25519 key fingerprint is SHA256:AxxC3J7swl/yj1oUjJCe8+QNhhwbAVDTitCYmDEr3jY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '51.20.184.180' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-2-62 ~]$
```

File Transfer

Project files were transferred from local machine to EC2.



```
C:\Users\MUB>cd C:\xampp\htdocs

C:\xampp\htdocs>scp -i "C:\Users\MUB\Downloads\ScholarHub-Key.pem" -r scholarship_portal ec2-user@51.20.184.180:/home/ec2-user/
add_scholarship.php                                                    100% 3098    18.6KB/s   00:00
approve.php                                                            100%  207     1.3KB/s   00:00
dashboard.php                                                          100% 3593    21.8KB/s   00:00
edit_scholarship.php                                                   100% 2598    16.3KB/s   00:00
logout.php                                                             100%    0     0.0KB/s   00:00
manage_scholarships.php                                                100% 3103    18.4KB/s   00:00
New Text Document.txt                                                  100%    0     0.0KB/s   00:00
reject.php                                                             100%  207     1.3KB/s   00:00
view_applications.php                                                  100% 2322    12.6KB/s   00:00
view_students.php                                                      100% 1539     9.5KB/s   00:00
db.php                                                                 100%  248     1.5KB/s   00:00
footer.php                                                             100%  118     0.7KB/s   00:00
header.php                                                             100% 1661    10.0KB/s   00:00
index.php                                                              100% 1926    12.0KB/s   00:00
login.php                                                              100% 2493    15.0KB/s   00:00
logout.php                                                             100%  118     0.7KB/s   00:00
New Text Document.txt                                                  100%    0     0.0KB/s   00:00
register.php                                                           100% 2668    16.1KB/s   00:00
Admin Dashboard.jpg                                                    100%  133KB 276.6KB/s   00:00
Apply scholarship Page.jpg                                             100%   72KB 216.9KB/s   00:00
Avaliable Scholarship Page.jpg                                         100%  109KB 209.3KB/s   00:00
Amazon RDS Connectivity and Security Configuration.jpg                 100%  184KB 539.1KB/s   00:00
Amazon RDS MySQL Connectivity and Security Configuration.jpg           100%  147KB 417.6KB/s   00:00
Amazon RDS MySQL Database Creation Configuration.jpg                   100%  136KB 405.8KB/s   00:00
Apache HTTP Server Running on Amazon EC2 Instance.jpg                  100%  227KB 644.1KB/s   00:00
database created.jpg                                                   100%   51KB 159.2KB/s   00:00
EC2  to  RDS MySQL Connection Successful.jpg                           100%   73KB 223.6KB/s   00:00
```

Apache Installation

Apache web server was installed and configured.

```
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-06-13 17:34:17 UTC; 18s ago
       Docs: man:httpd.service(8)
   Main PID: 28508 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1014)
     Memory: 13.4M
        CPU: 99ms
     CGroup: /system.slice/httpd.service
             ├─28508 /usr/sbin/httpd -DFOREGROUND
             ├─28560 /usr/sbin/httpd -DFOREGROUND
             ├─28564 /usr/sbin/httpd -DFOREGROUND
             ├─28565 /usr/sbin/httpd -DFOREGROUND
             └─28572 /usr/sbin/httpd -DFOREGROUND

Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal httpd[28508]: Server configured, listening on: port 80
[ec2-user@ip-10-0-2-62 ~]$
```

PHP Installation

PHP runtime environment was installed.

```
[ec2-user@ip-10-0-2-62 ~]$ php -v
PHP 8.5.6 (cli) (built: May  5 2026 21:19:36) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.6, Copyright (c), by Zend Technologies
[ec2-user@ip-10-0-2-62 ~]$
```

MySQL Client Installation

MySQL client tools were installed for database connectivity.

```
  Verifying        : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64              mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64                    mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-10-0-2-62 ~]$ mysql --version
mysql  Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using  EditLine wrapper
[ec2-user@ip-10-0-2-62 ~]$
```

Application Deployment:

The ScholarHub application was successfully deployed on an Amazon EC2 instance running Apache HTTP Server and PHP. The application was migrated from the local XAMPP environment to the cloud environment, enabling remote accessibility through a public IP address.

Component	Technology Used
Web Server	Apache HTTP Server
Compute Service	AWS EC2
Database	Amazon RDS MySQL
Storage	Amazon S3
CDN	Amazon CloudFront
Monitoring	AWS CloudWatch
Cost Monitoring	AWS Budgets
IaC	Terraform
Application Language	PHP
Frontend	HTML, CSS, JavaScript

Live Deployment URL

Application URL:

http://51.20.184.180

or

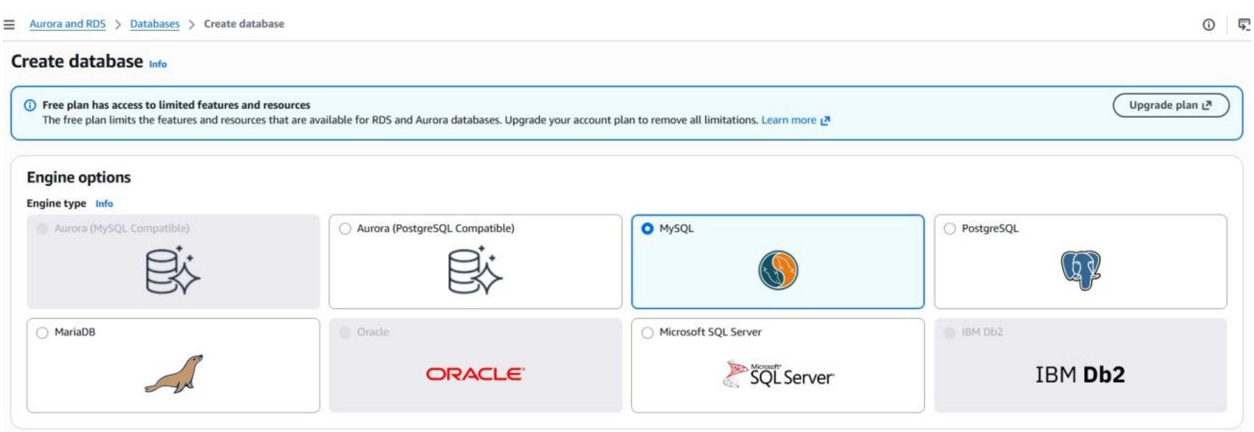
http://ec2-51-20-184-180.eu-north-1.compute.amazonaws.com

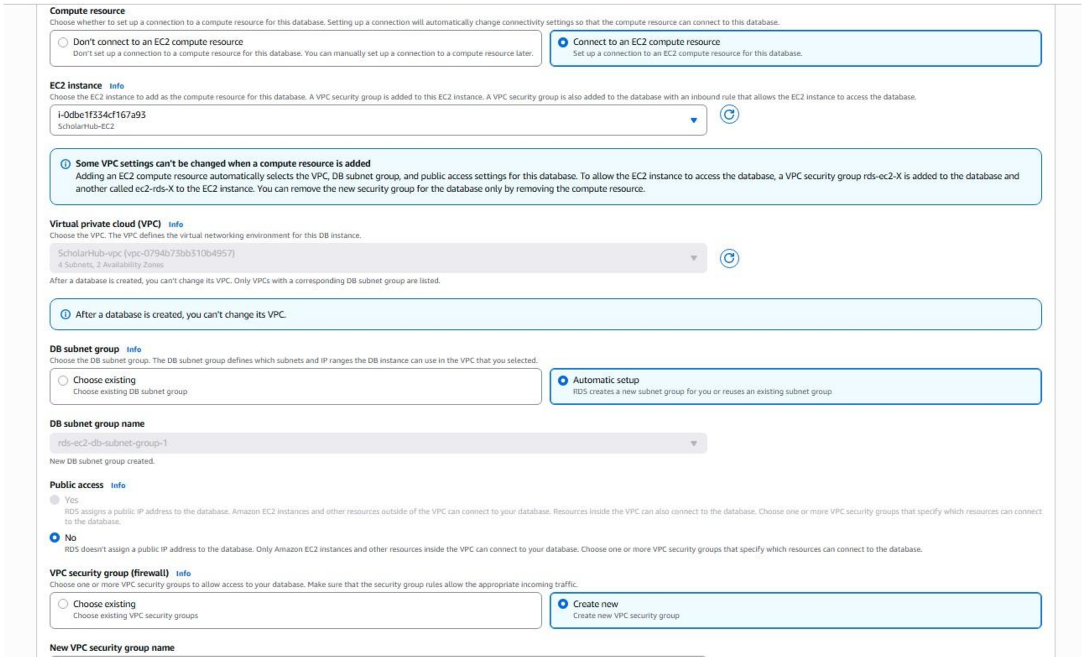
Database and Storage Layer

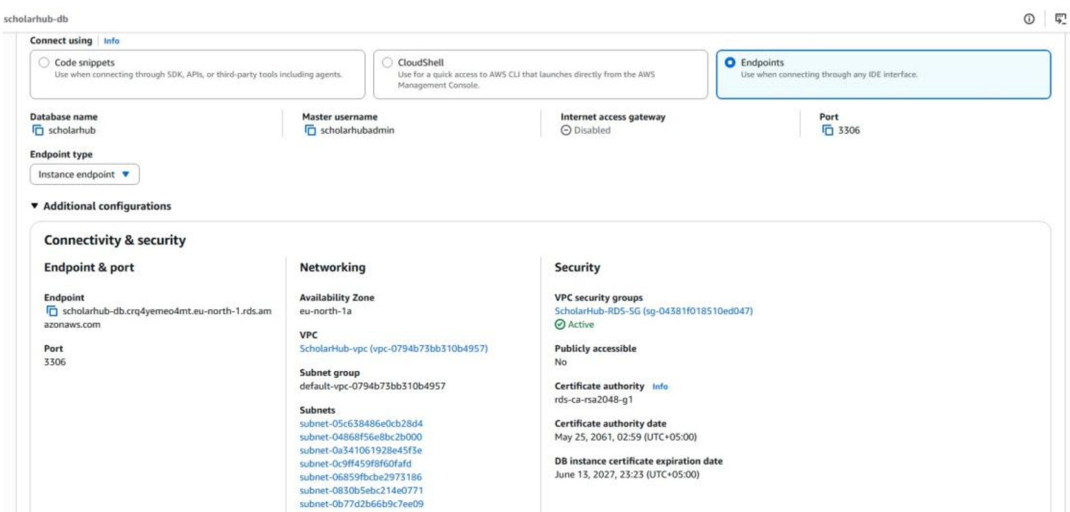
Amazon RDS

Amazon RDS MySQL was used as the managed database service.

- Automated backups
- Managed maintenance
- Improved reliability
- High availability support

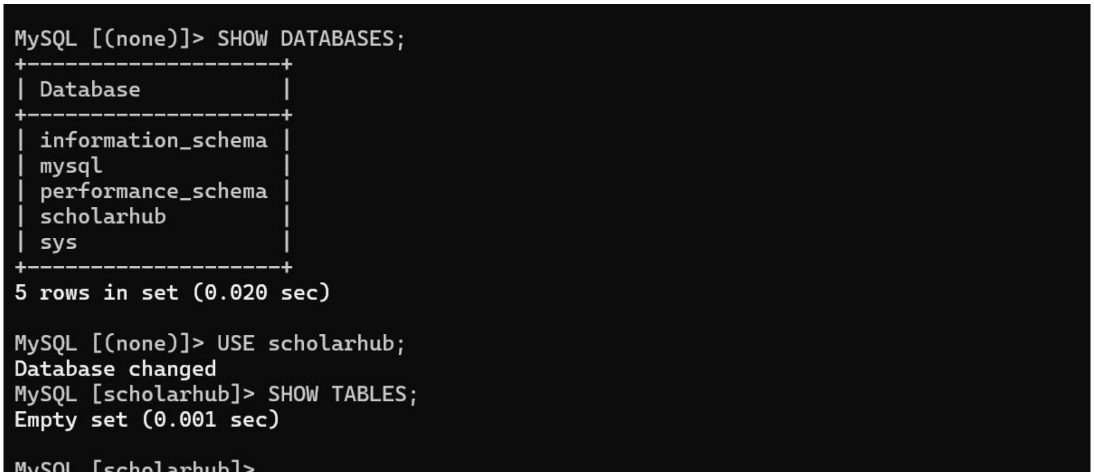




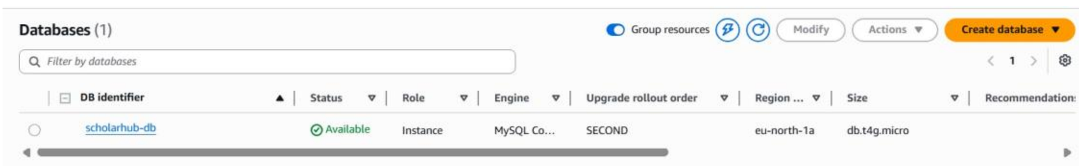


Ec2 to Rds mysql connection:

```
[ec2-user@ip-10-0-2-62 ~]$ mysql -h scholarhub-db.crq4yemeo4mt.eu-north-1.rds.amazonaws.com -P 3306 -u scholarhubadmin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```



```
MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| scholarhub         |
| sys                |
+--------------------+
5 rows in set (0.020 sec)

MySQL [(none)]> USE scholarhub;
Database changed
MySQL [scholarhub]> SHOW TABLES;
Empty set (0.001 sec)

MySQL [scholarhub]>
```



Automated Backups

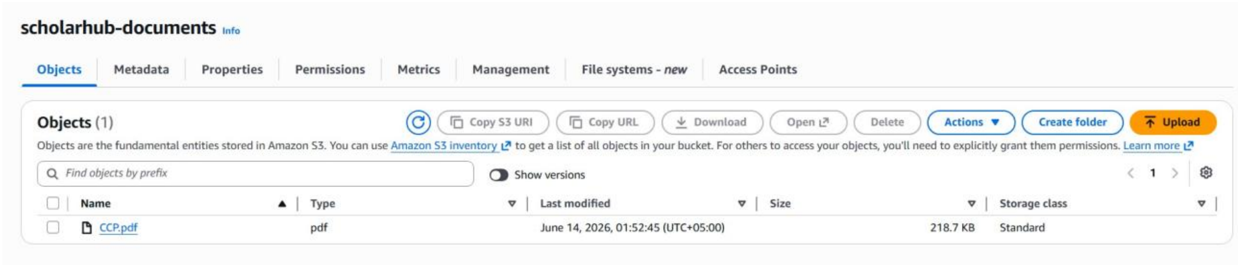
Automated backup retention was enabled.

Amazon S3:

Student supporting documents are stored in Amazon S3.

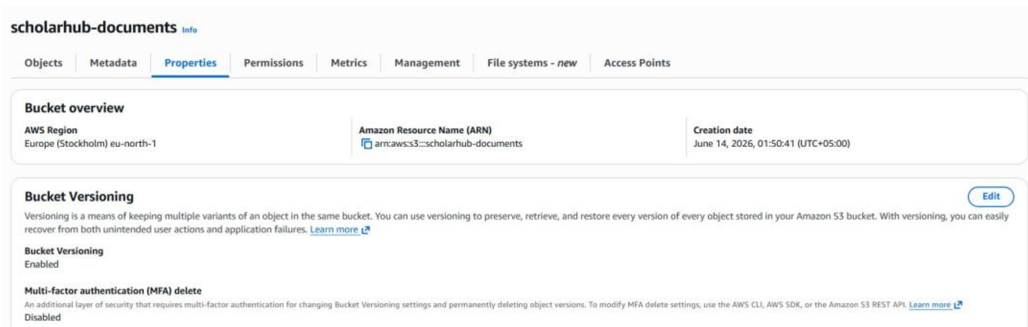
Benefits:

- Durable storage
- Scalability
- Low cost
- Secure file management

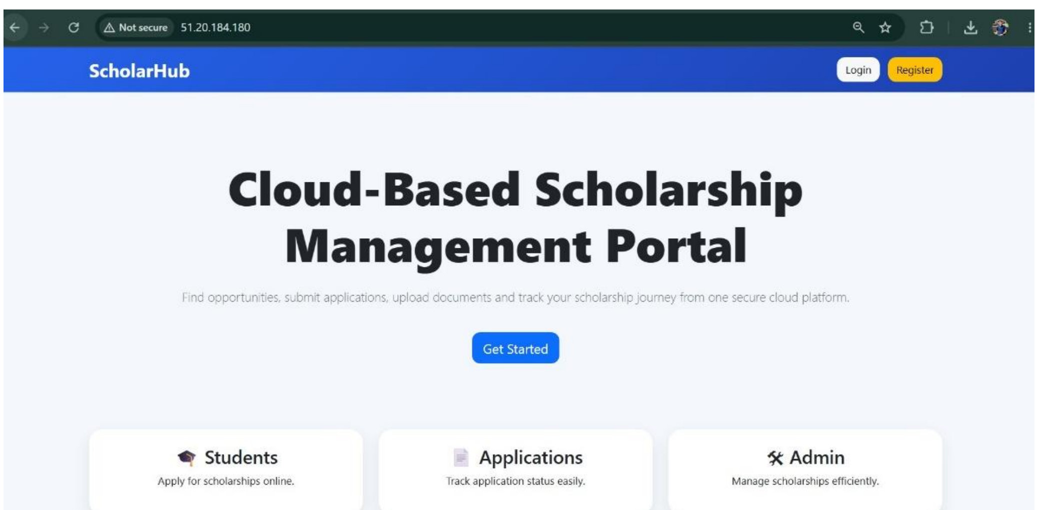


Bucket Versioning:

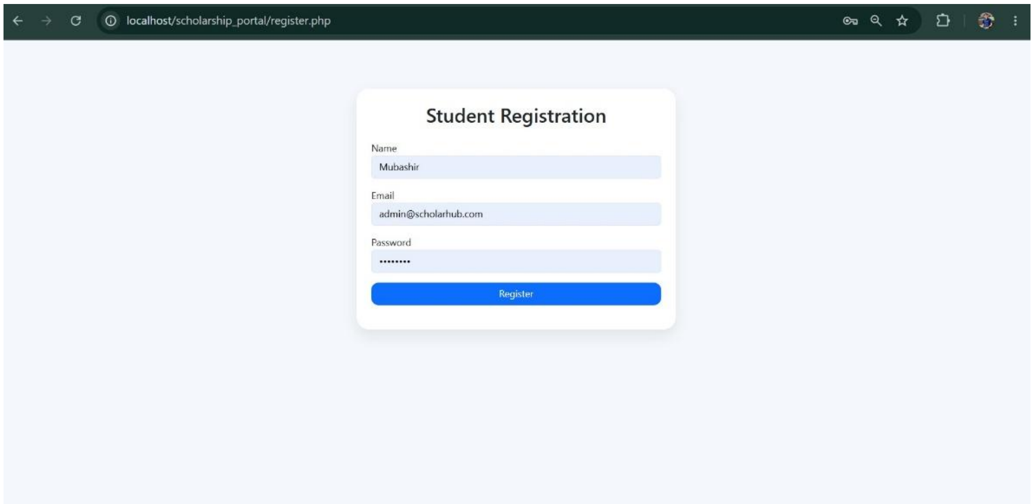
Versioning was enabled to protect against accidental deletion.



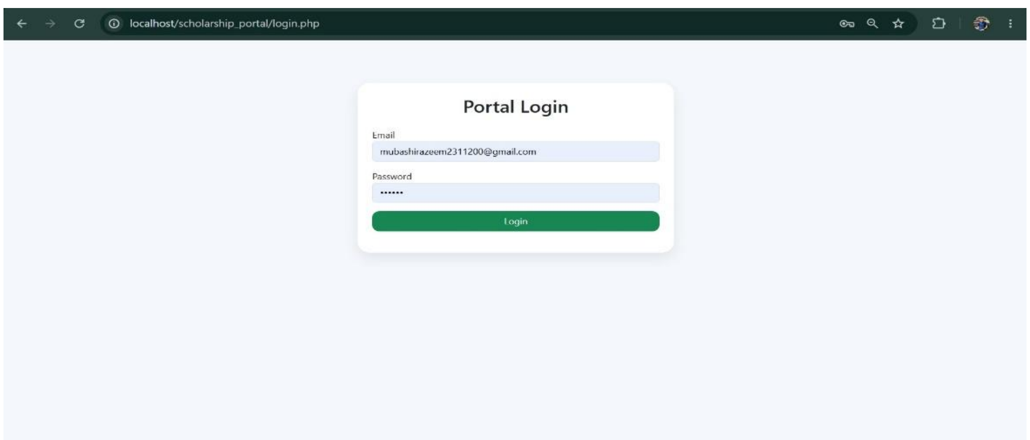
Frontend and Application Features



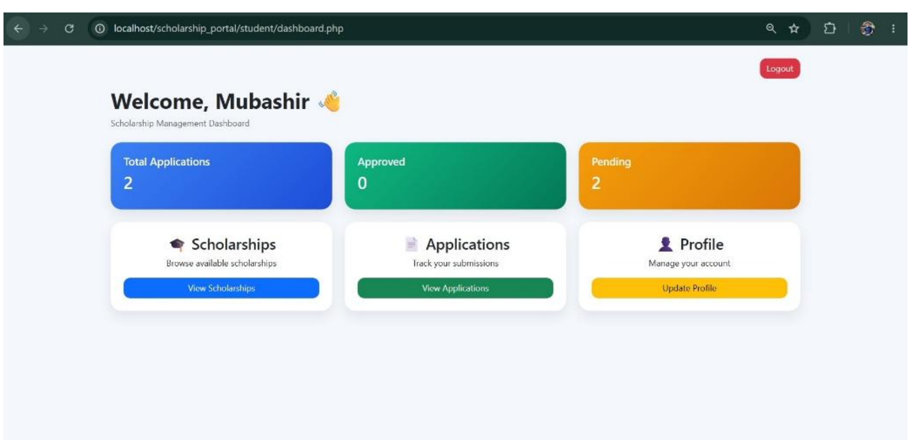
Registration Page:



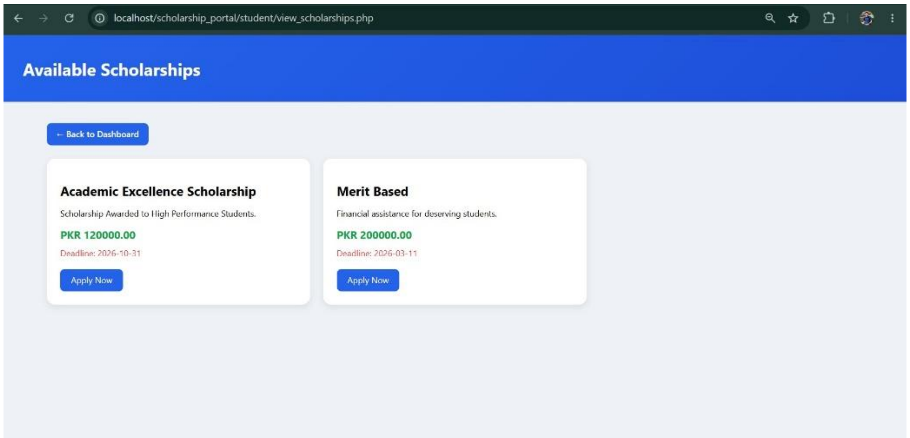
Login Page:



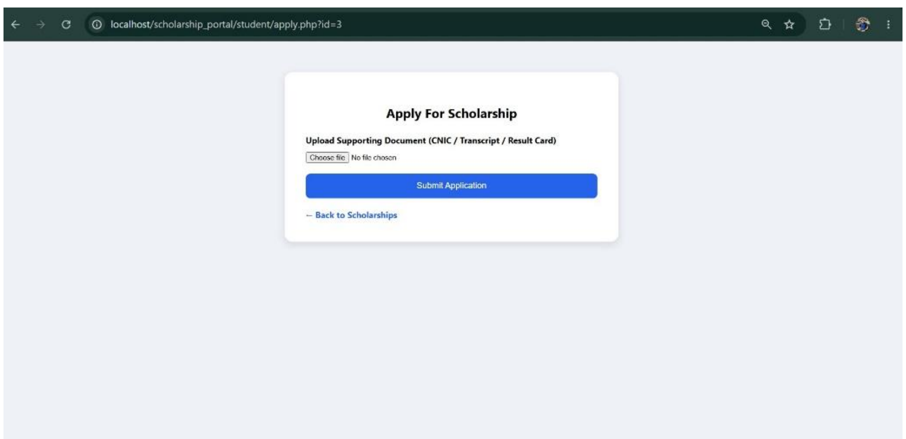
Student Dashboard:



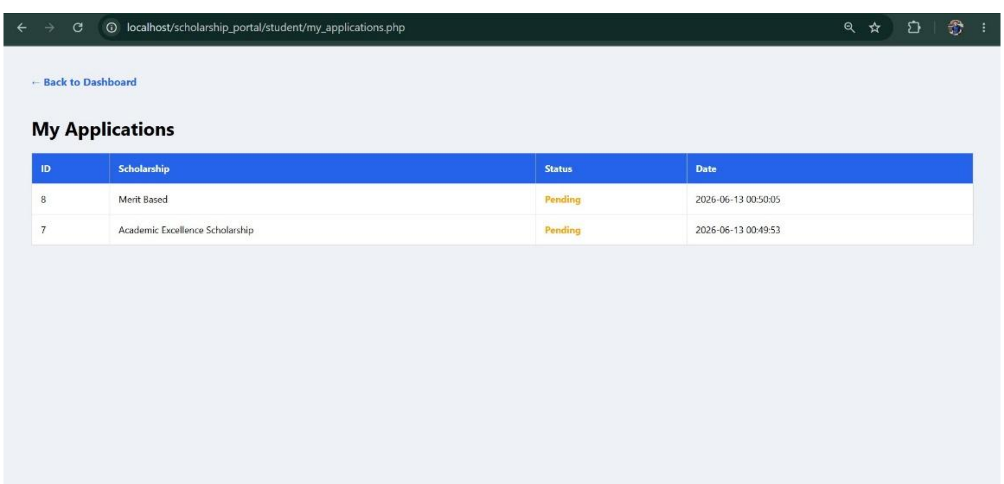
Available Scholarships:



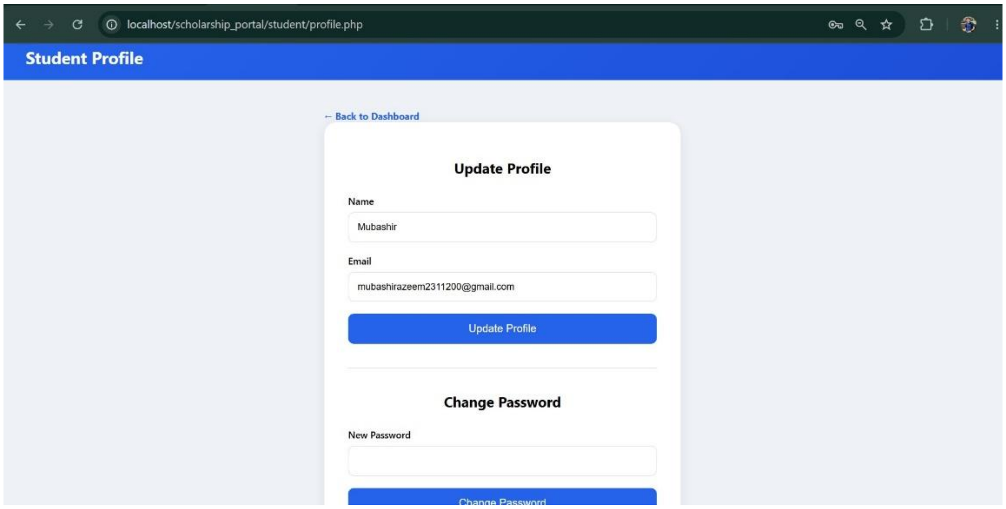
Apply Scholarship:



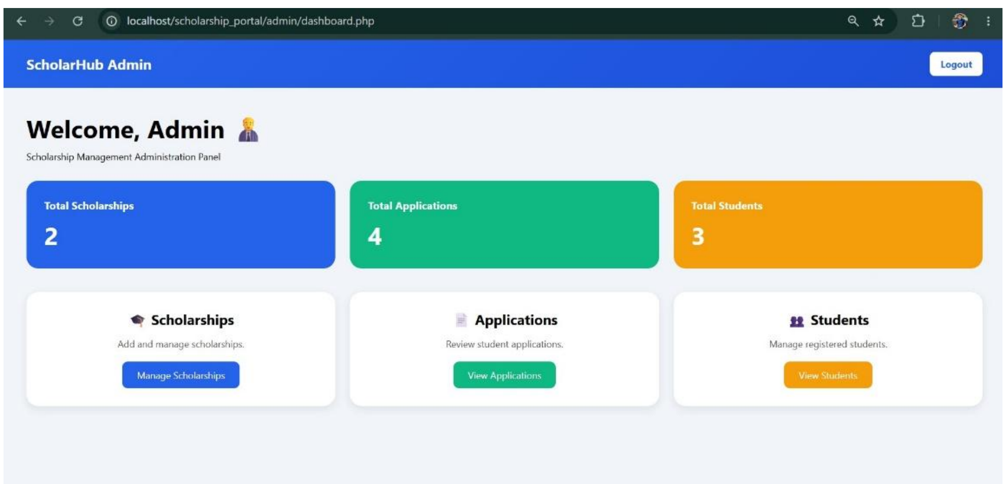
My Applications:



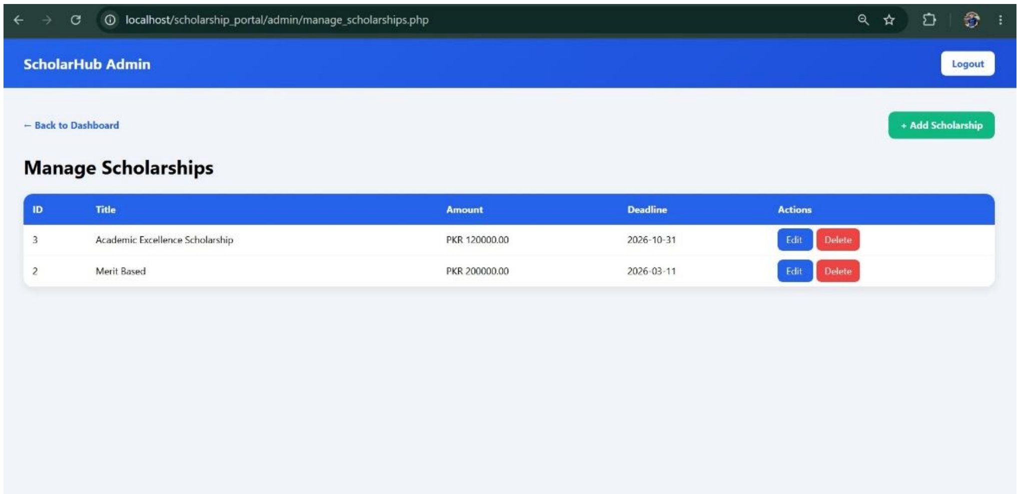
Profile Page:



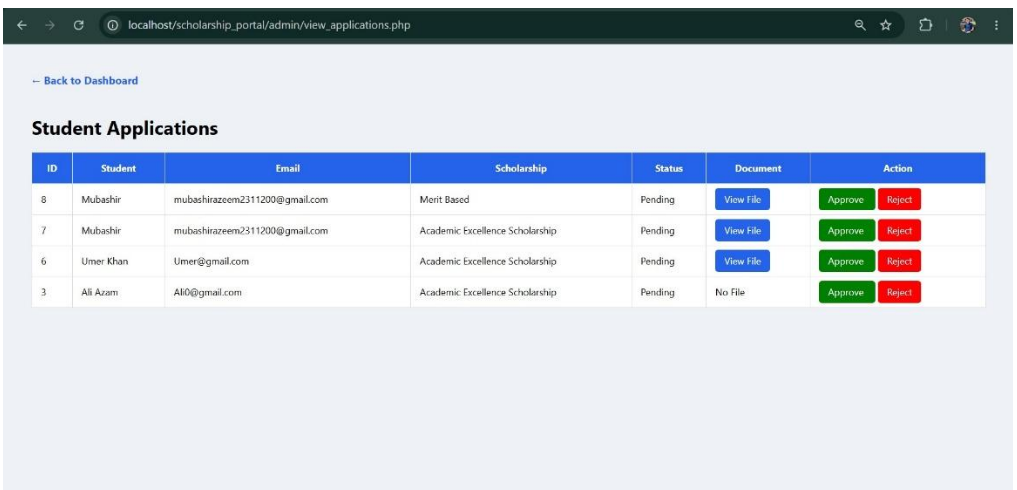
Admin Dashboard:



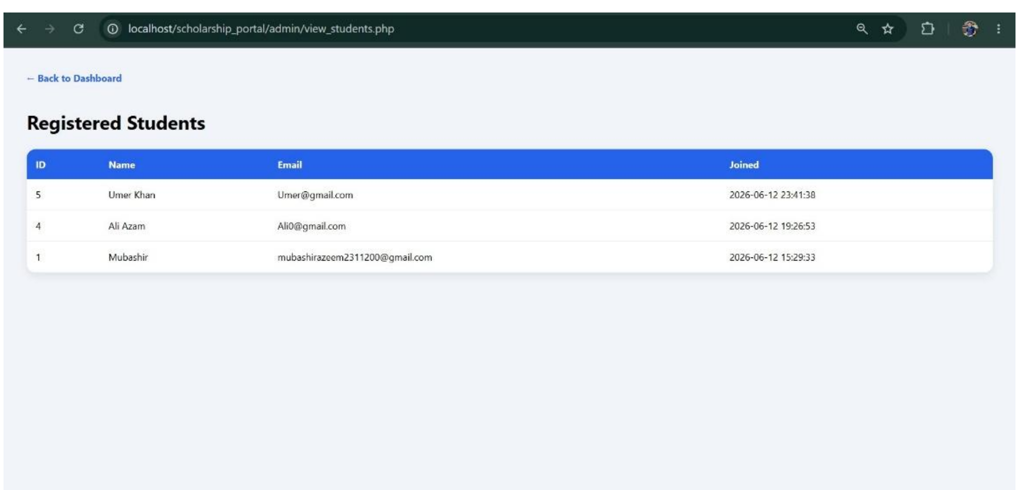
Manage Scholarships:



View Applications



View Students



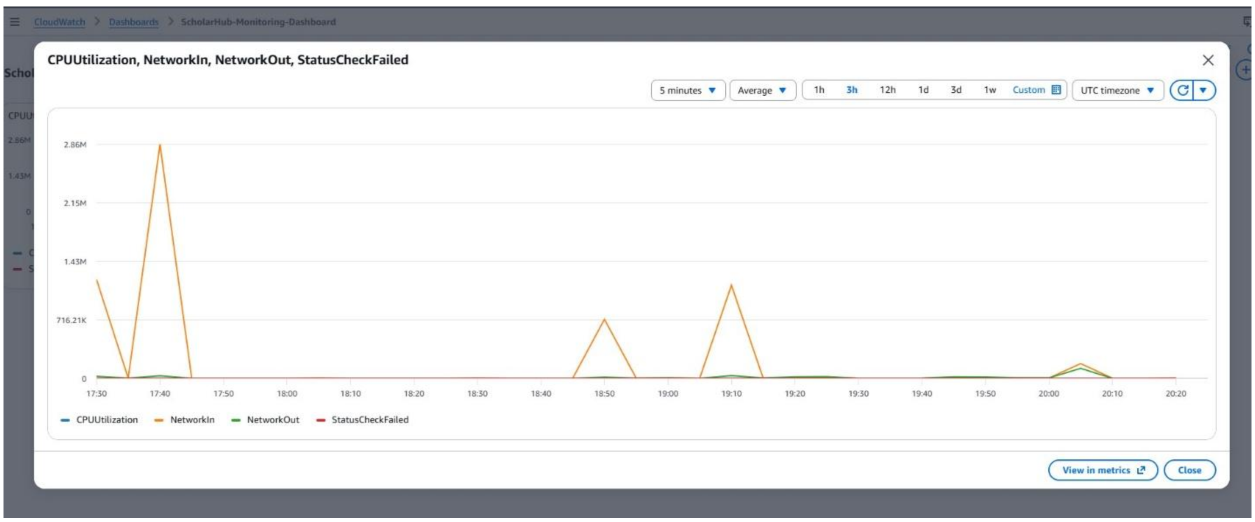
Monitoring and Logging:

AWS CloudWatch

CloudWatch was configured to monitor system health.

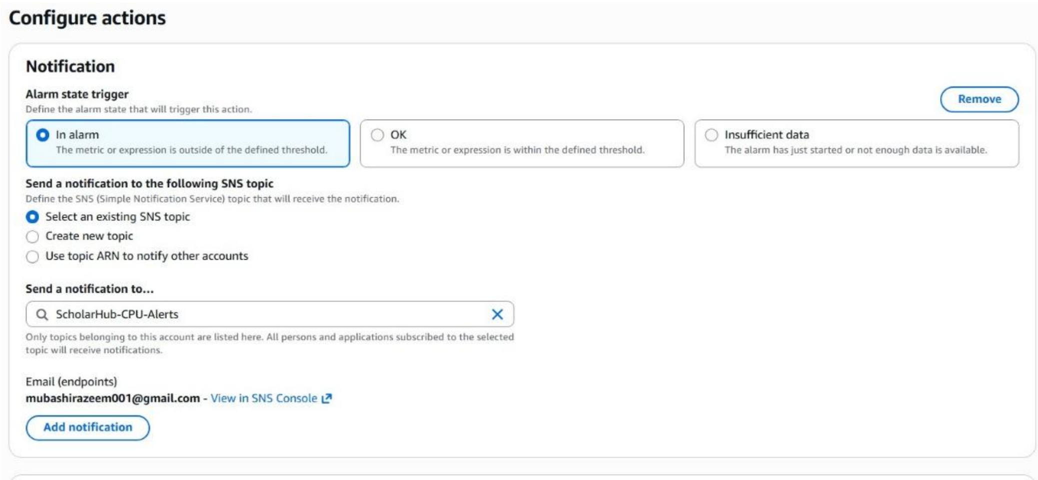
Metrics monitored:

- CPU Utilization
- Network In
- Network Out
- Status Checks



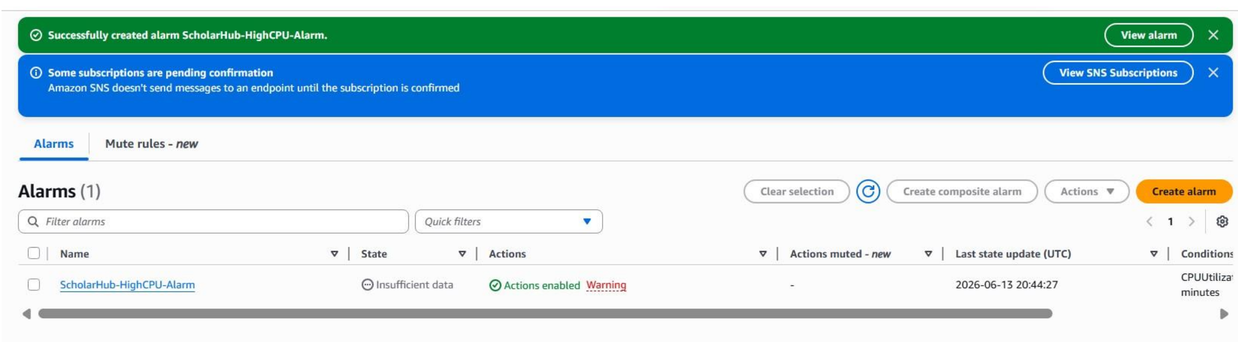
CloudWatch Alarm:

An alarm was configured to trigger when CPU usage exceeds 80%.



Purpose of CloudWatch

CloudWatch continuously collects performance metrics and helps administrators identify issues before they affect application availability.



Cost Monitoring:

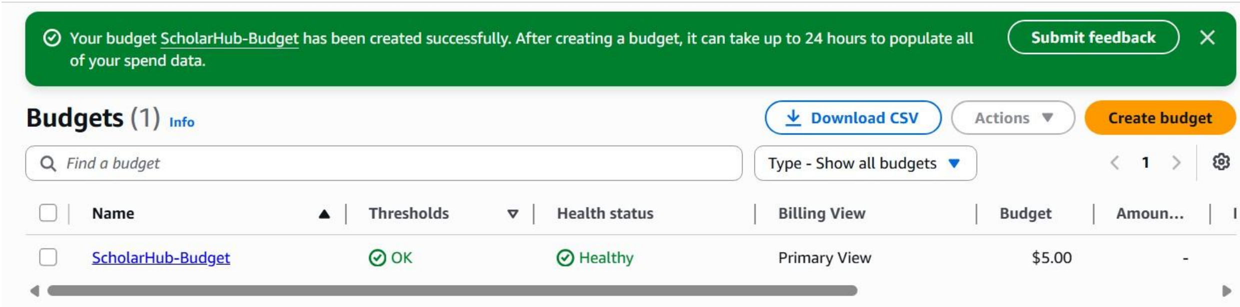
AWS Budgets was configured.

Budget Limit:

- $5 Monthly Budget

Benefits:

- Cost visibility
- Spending alerts
- Preventing unexpected charges

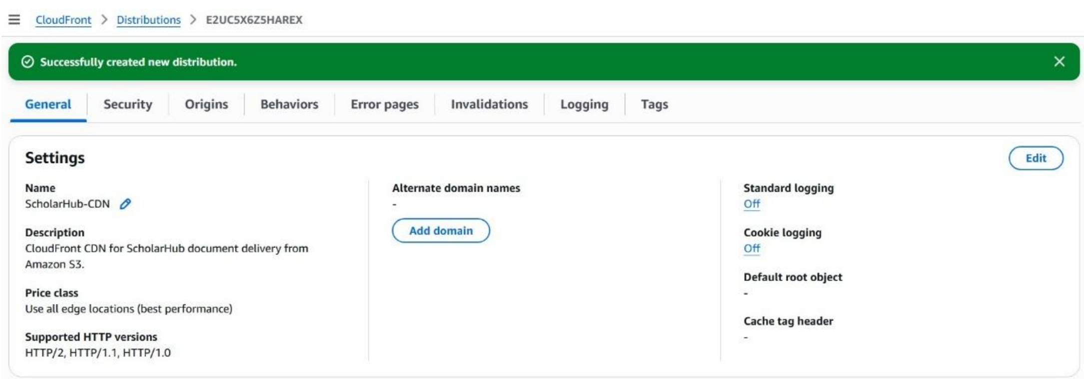


CloudFront CDN:

Amazon CloudFront was configured to distribute files stored in S3.

Benefits:

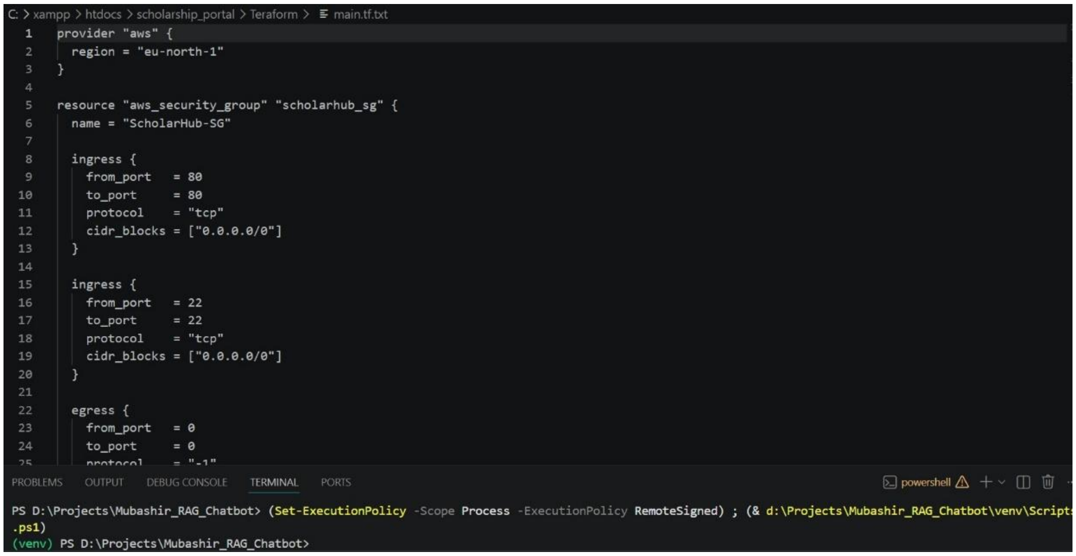
- Faster content delivery
- Global edge caching
- Reduced latency
- Improved user experience



Infrastructure as Code (Terraform):

Terraform was used to demonstrate Infrastructure as Code (IaC).

- Automated resource creation
- Reproducible infrastructure
- Faster deployment
- Reduced manual configuration



Results and Discussion:

The system was successfully deployed on AWS and all major functionalities operated correctly.

User Flow 1:

Student Registration to **Login** to **Dashboard**

User Flow 2:

Browse Scholarships to **Apply** to **Upload Documents**

User Flow 3:

Admin Login to **Manage Scholarships** to **View Applications**

Cost Analysis and Security Considerations:

Cost Optimization:

The ScholarHub portal was deployed using AWS Free Tier services to minimize operational costs while maintaining required cloud functionality.

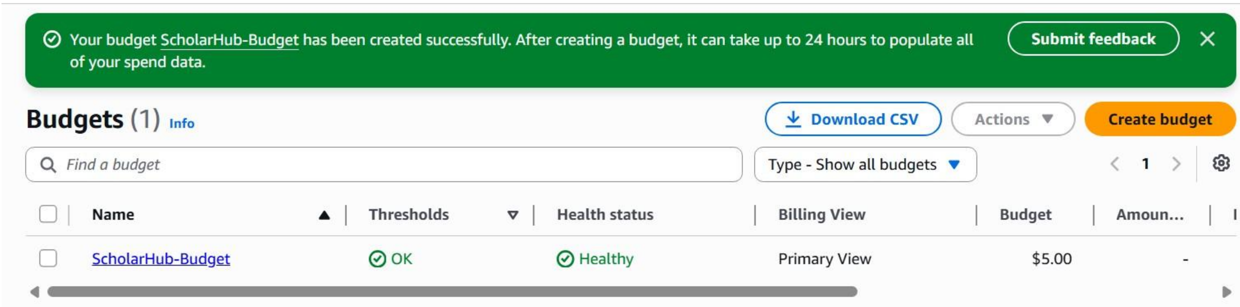
Cost Optimization Techniques

- AWS Free Tier utilized
- Single EC2 instance deployment
- Amazon RDS MySQL Free Tier instance
- AWS Budget alerts configured

- CloudWatch monitoring enabled
- CloudFront caching enabled
- Amazon S3 used for low-cost storage

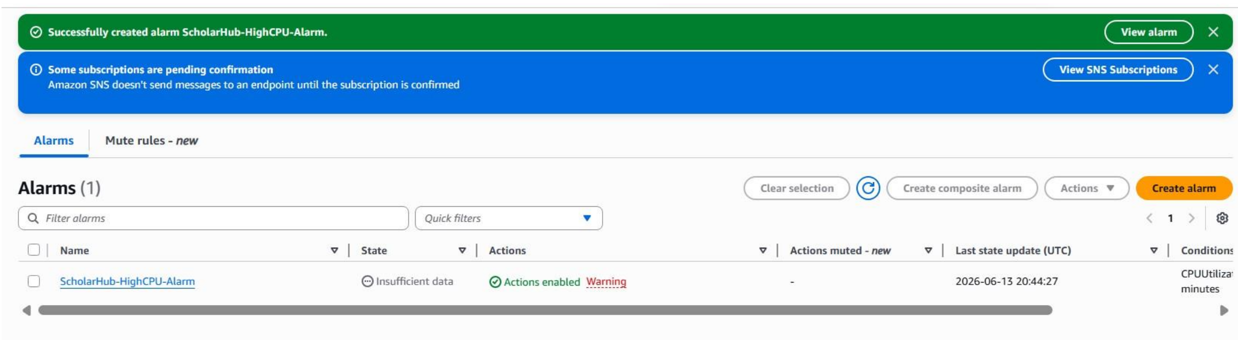
AWS Budget Monitoring

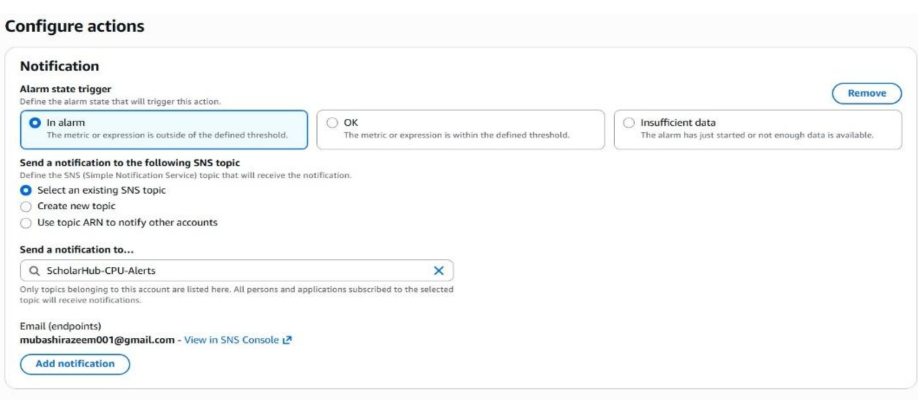
AWS Budgets was configured to monitor monthly cloud spending and generate alerts when the budget threshold is reached.



Cloud Monitoring

AWS CloudWatch was configured to monitor system performance including CPU utilization, network traffic, and instance health.





Conclusion:

The Cloud-Based Scholarship Management Portal was successfully designed, developed, and deployed on AWS. The project demonstrates the practical implementation of cloud computing technologies including Amazon EC2, Amazon RDS, Amazon S3, Amazon CloudFront, Amazon CloudWatch, AWS Budgets, and Terraform. The system provides a secure, scalable, and cost-effective solution for managing scholarship applications while fulfilling modern cloud deployment requirements.

The application was successfully hosted on AWS EC2 and integrated with Amazon RDS MySQL for database management. Additional cloud services such as S3 for document storage, CloudFront for content delivery, CloudWatch for monitoring, AWS Budgets for cost control, and Terraform for Infrastructure as Code (IaC) were incorporated to enhance functionality, performance, security, and maintainability. The project successfully demonstrated real-world cloud deployment practices and achieved all major project objectives.

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